Journal of Sports Sciences, December 2007; 25(14): 1557 – 1566

Routledge
Taylor & Francis Group

The role of a short post-lunch nap in improving cognitive, motor, and sprint performance in participants with partial sleep deprivation

J. WATERHOUSE, G. ATKINSON, B. EDWARDS, & T. REILLY

Research Institute for Sport and Exercise Sciences, Liverpool John Moores University, Liverpool, UK

(Accepted 26 January 2007)

Abstract
The aim of this study was to determine the effects of a post-lunch nap on subjective alertness and performance following partial sleep loss. Ten healthy males (mean age 23.3 years, $s = 3.4$) either napped or sat quietly from 13:00 to 13:30 h after a night of shortened sleep (sleep 23:00 – 03:00 h only). Thirty minutes after the afternoon nap or control (no-nap) condition, alertness, short-term memory, intra-aural temperature, heart rate, choice reaction time, grip strength, and times for 2-m and 20-m sprints were recorded. The afternoon nap lowered heart rate and intra-aural temperature. Alertness, sleepiness, short-term memory, and accuracy at the 8-choice reaction time test were improved by napping ($P < 0.05$), but mean reaction times and grip strength were not affected ($P > 0.05$). Sprint times were improved. Mean time for the 2-m sprints fell from 1.060 s ($s_{\bar{x}} = 0.018$) to 1.019 s ($s_{\bar{x}} = 0.019$) ($P = 0.031$ paired t-test); mean time for the 20-m sprints fell from 3.971 s ($s_{\bar{x}} = 0.054$) to 3.878 s ($s_{\bar{x}} = 0.047$) ($P = 0.013$). These results indicate that a post-lunch nap improves alertness and aspects of mental and physical performance following partial sleep loss, and have implications for athletes with restricted sleep during training or before competition.

Keywords: *Sleep loss, athletic performance, fatigue, daytime sleep*

Introduction

Sleep loss affects mood and performance at a variety of mental tasks. In general, there is a fall in mood and performance after as little as 2 – 3 h sleep loss, although the size of the decrement depends on the type of mental task. Decrements are more marked when repetitive and boring tasks are involved. [For some recent reviews, see Dinges (1995), Folkard (1990), Rosekind, Flower, Gregory, & Jung (2005), Van Dongen & Dinges (2005), and Waterhouse, Minors, Åkerstedt, Reilly, & Atkinson (2001).] Short sleeps (naps) ameliorate these decrements in performance and negative effects on mood (Van Dongen & Dinges, 2005), although, unless the nap is short and devoid of slow-wave sleep, mental performance and mood are temporarily worsened until the individual has woken up fully. This transient effect is known as ''sleep inertia'', and its effects wear off within the first hour after waking up (Brooks & Lack, 2005; Naitoh, 1981; Stampi Mullington, Rivers, Campos, & Broughton, 1990).

Several mathematical simulations of mental performance have appeared, generally incorporating separate effects due to circadian phase, time since waking, and sleep history (Åkerstedt & Folkard, 1995; Krauchi, Cajochen, & Wirz-Justice, 2005; Neri, 2004; Spencer, 1987; Waterhouse *et al.*, 2001), and these provide a rationale for giving advice on ways to combat the negative effects due to disturbed sleep, a period of extended wakefulness, or an abnormal phasing of circadian rhythms.

With regard to physical performance, there is less information about the effects of limited sleep loss (Leger Metlaine, & Choudat, 2005; Smith & Reilly, 2005) and of any effects that naps might have. Such information would be valuable when there is a requirement for physical activity in individuals who are suffering from some degree of sleep loss, as during night work or after time-zone transitions (Reilly, Edwards, Waterhouse, & Atkinson, 2005; Reilly, Atkinson, & Waterhouse, 1997). There is also a potential relevance to sportspersons, since even small decrements in physical performance might separate a winning team or individual from another whose performance is only slightly below peak values, and a degree of sleep loss is often present in them (Leger *et al.*, 2005) and possibly contributes to

Correspondence: J. Waterhouse, Research Institute for Sport and Exercise Sciences, Liverpool John Moores University, Henry Cotton Campus, 15 – 21 Webster Street, Liverpool L3 2ET, UK. E-mail: waterhouseathome@hotmail.com

ISSN 0264-0414 print/ISSN 1466-447X online © 2007 Taylor & Francis
DOI: 10.1080/02640410701244983

"underperformance syndrome". Sleep loss might arise from the residual effects of disturbed circadian rhythms following a time-zone transition, but it might also arise when the event is taking place in the home time zone; sleep might be more disturbed than normal due to having to sleep in unfamiliar surroundings, to apprehension the night before an important competition, or to noise from sports fans, for example. In addition, travel to a meeting might involve getting up early in the morning. Partial sleep loss is also likely to be quite common in some domestic circumstances – when there is a newborn child, for example – when the disturbance might last several weeks and might affect performance during routine training schedules the following day.

Some research (for reviews, see Reilly *et al.*, 2005; Smith & Reilly, 2005) has investigated physical performance after a time-zone transition and reported transient decrements in performance. However, this circumstance does not make it possible to distinguish between the effects of abnormally phased circadian rhythms (due to the time-zone transition) and sleep loss *per se*. Meney and colleagues (Meney, Waterhouse, Atkinson, Reilly, & Davenne, 1998) studied the effects of one night's sleep loss in individuals continuing to live in the same time zone, so removing effects due to performing physical activity at a changed circadian phase. They measured core temperature, mood, and several measures of physical performance (handgrip, leg strength, back strength, and self-chosen intensity of exercise). Changes to physical performance measures were generally non-significant, even though mood was negatively affected. By contrast, more extended periods of sleep loss, as in an exhibition soccer marathon, for example, resulted in a fall on successive days in the percentage of time the individuals were actively engaged in playing soccer (Reilly & Walsh, 1981). The exact relevance of these studies to the effect of only limited sleep loss on sporting performance is unclear; the former study measured aspects of physical performance that are only surrogates for sporting performance itself, and the latter, while measuring a sporting task, involved excessive sleep loss. Moreover, neither study investigated the effects of a nap on any deterioration in performance that might be present.

Of relevance to athletes would be an investigation into the effects of a nap on athletic performance in the presence of smaller amounts of sleep loss as might commonly occur (see above) during training schedules or before competition. In practice, the timing of a nap is often determined by the individual's work schedule; a nap around lunchtime is not only often easier to incorporate into a daily schedule but also it is at a time when there is a transient fall in alertness and increased ability to nap (Brooks & Lack, 2005; Folkard, 1990; Reilly *et al.*, 1997).

The present study was undertaken to address directly some of these issues. We assessed the effectiveness of a lunchtime nap in ameliorating effects on a sporting task caused by a limited amount of loss of sleep during the previous night in participants otherwise living regular and conventional sleep – wake schedules.

Methods

Participants

Ten males (mean age 23.3 years, $s = 3.4$) participated in the study. From questioning the participants, it was established that none experienced any sleep complaints, and all were in good health and slept at conventional hours (approximately 23:00 – 07:00 h; see below). All participants were right hand dominant. None of them had an understanding of the possible effects of sleep loss and naps.

Ethics

The experimental procedures were in accord with university guidelines and passed by the human ethics committee of the institution. The procedures were explained in full to all participants and any questions about the protocol were answered. They were required to sign consent forms in the presence of independent witnesses. Participants were informed that they could withdraw from the experiment at any time, although none did so.

Protocol

Before the main experimental sessions, all participants attended two habituation sessions, one week apart, in the experimental laboratory. Participants answered all the questionnaires, practised all the tests, and had all the measuring equipment attached (for details, see below). During one of the habituation sessions, participants were shown the quiet room and bed in which they would have an opportunity to take their nap.

For the main experiment, a crossover experimental protocol was used, with participants being tested on two separate occasions at least one week apart. Both arms of the study were undertaken after a night of partial sleep loss. To achieve this, participants were instructed to sleep 4 h less than normal. This required the participants to retire the night before a test session at their normal time (between 22:30 and 23:30 h for the group as a whole) and to rise 4 h earlier than normal (between 02:30 and 03:30 h for the group as a whole). After this early rising,

participants were free to choose their activities. In practice, after switching on the light, all watched television or listened to music.

Participants were asked to refrain from any vigorous physical activity in the 24 h prior to each of the experimental sessions, during which time alcoholic or caffeine-containing beverages were prohibited. No food or drink was allowed in the 4 h beforehand.

Participants arrived at the experimental laboratory at 12:30 h, when they were informed (before the first session) as to whether it was the nap or no-nap condition. Participants were randomly assigned to either the "nap" or "no-nap" condition for the first experimental session. Participants in the nap condition entered a room that was conducive to sleep (comfortably warm, dimly lit, and quiet) at about 12:45 h, and got into bed. After being allowed a few minutes to become accustomed to their surroundings and to become comfortable, they were allowed 30 min to nap (13:00 – 13:30 h). This length of time for a nap was chosen as it would often be possible for it to be incorporated into a daily routine. After this time, participants were woken, if they were asleep, and allowed 30 min to overcome any sleep inertia that might have existed. In the no-nap condition, the same period (12:45 – 14:00 h) was spent awake, the participants reading quietly. One of the authors ensured that this protocol was adhered to by all participants.

At 14:00 h, all participants recorded on analogue scales (see below) their sleepiness and alertness, together with (if in the nap condition) the quality of their nap; they then started the tests. These consisted of four components, performed in the following order:

- short-term memory test;
- visual choice reaction time test;
- handgrip strength;
- sprint running times over 2 and 20 m.

Performance measurements

Subjective sleepiness, alertness, and sleep quality. Measurements of these variables were made using visual analogue scales based on the Stanford Sleepiness Scale (Hoddes Zarcone, Smythe, Phillips, & Dement, 1973). "Sleepiness" was measured on a scale of 1 – 7, where 1 indicated "high activeness" and 7 an equally "high tiredness". Alertness was measured on a scale of 0 – 10, where 0 indicated "extreme drowsiness" and 10 "extreme alertness". Subjective ratings of sleep quality during the nap were made using a scale of 0 – 10, where 0 indicated "no sleep", 5 indicated "some sleep with some interruptions", and 10 indicated "uninterrupted, deep sleep throughout".

Short-term memory test. The test used for short-term memory was a digit span test (Troy *et al.*, 2000). It had two components: digits-forwards and digits-backwards. With the digits-forwards component, the participant was read a series of random numbers, starting with a sequence of five digits, and asked to repeat it to the experimenter in the same order in which it had been presented. If the participant was successful, the number of digits was raised by one, and the test continued; whenever the participant was unsuccessful, the test continued but with a new sequence that was the same length as the previous one. In total, 10 sequences were given, the length of the last sequence that had been repeated successfully being taken as the score for that test.

The digits-backwards component was performed in the same way, except that the first sequence was 4 digits in length and the participant was required to repeat it to the experimenter in the reverse order. Rules for increasing the length of the sequence were the same as with the digits-forwards component. Again, 10 tests were given and the length of the last sequence that had been successfully repeated was the score for the test.

Visual choice reaction time test. The version used investigated prepared reaction times when the number of choices was one, two, four, and eight (Hick, 1952). It was run on a desktop PC. A warning square was presented in the bottom centre of the screen, followed shortly by a stimulus key, "+", again at the bottom of the screen. The required response was to press the keyboard key under the "+" sign as soon as possible. With the one-choice test, the stimulus was always at the same position, the centre of the screen, and the test amounted to a measurement of simple reaction time. With the two-choice test, the stimulus could appear at one of two symmetrically placed positions along the bottom of the screen. With the four- and eight-choice tests also, the possible positions were placed symmetrically along the bottom of the screen.

The test consisted of four trials, using one, two, four, and eight choices, respectively. There were eight stimuli in each trial. For each trial, both the mean response time (measured to 0.001 s) and the number of wrong key presses were recorded. The test took less than 10 min to complete.

Handgrip strength. Grip strength for both the left (non-dominant) and right (dominant) hands was recorded using a dynamometer (T.K.K. 5101, Japan). The participant gripped the dynamometer in a raised position above his head and then lowered the dynamometer until it had reached waist height. Grip strength was recorded from left and right hands alternately. Three values from each

hand were recorded, each of these being used in the analysis.

Sprint running times. Three sprints were performed, with a 1-min rest between sprints. Sprint performance times were recorded from 2-m and 20-m points, using a set of standard battery timing lights (PGS Timing System MK II; SICK Optic Electronic, Germany). The participants were placed behind the starting line and used a standard starting position with blocks. The sprint was initiated by an audio signal that came from behind the participant, so mimicking race conditions (Buckolz & Vigars, 1987). The times taken to cover 2 m and 20 m were recorded for each of the three sprints, and the three sprints were analysed separately. The accuracy of the timing mechanism was 0.001 s.

Other measurements

Heart rate was recorded four times (Polar Electro heart rate monitor, Favor, Finland): before (13:00 h) and after (13:30 h) the times allocated to the nap (or to quiet reading in the no-nap condition), and before (14:00 h) and after (about 14:30 h) the battery of performance tests. Core body temperature was estimated from the external auditory meatus, intra-aural temperature (Genius 3000 A infrared thermometer: Sherwood, Davis, & Gosport, Hampshire, UK) at the same times.

Statistical analyses

Statistical analyses were performed with the Statistics Package for the Social Sciences (SPSS) (version 11). More details of the hypothesis tests are given at the appropriate points in the "Results" section.

Corrections for violations of sphericity (Greenhouse-Geisser) and multiple comparisons of differences within a factor (Bonferroni) were used as appropriate (Atkinson, 2001, 2002). In the text and tables, exact P-values are given. Statistical significance was set at $P < 0.05$. Values where $0.10 > P > 0.05$ have been reported as marginally significant and other values, where $P > 0.10$, have been reported as "not significant" (Zar, 1999, p. 83). Values for P of "0.000" given by the statistics package have been recorded as "<0.0005".

Results

Heart rate and temperature

Figure 1 (top) shows the mean heart rate at the different stages of the experiment. The fall produced by the nap condition (but not the no-nap condition) is clear, as are the rises produced by the tests in both conditions. We separated the data into those obtained before/after the nap and those before/after the tests, so as to focus on the effects of the nap and the tests, respectively. A two-factor analysis of variance

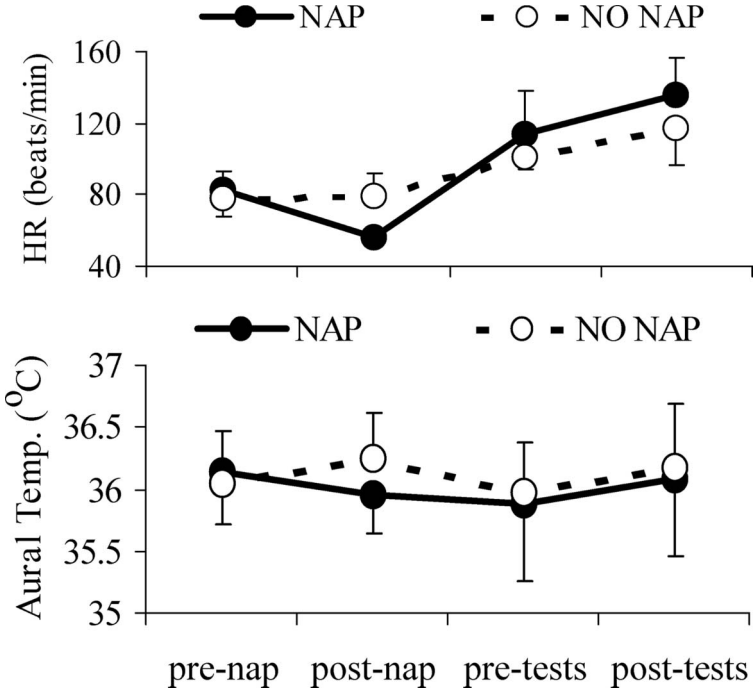

Figure 1. (Top) Heart rate in the nap and no-nap conditions. (Bottom) Aural temperature in the nap and no-nap conditions. Values are means and standard deviations. pre-nap/post-nap = before and after the time allocated to the nap; pre-tests/post-tests = before and after the test sessions.

(ANOVA), with repeated measures, was performed to test the main effect of nap and time and the interaction between these two factors (dependent variable = heart rate; fixed factors = nap/no-nap condition and pre-/post-nap or pre-/post-test session). The results of these analyses are shown in Table I. The significant interaction between the nap/no-nap condition and pre-/post-nap indicates that the nap condition alone produced a fall in heart rate. There was also a significant rise in both conditions following the test sessions.

Aural temperatures at the different stages of the experiment are shown in Figure 1 (bottom) and the associated statistics in Table I. Again, the significant interaction effect indicates that the nap condition alone produced a fall in aural temperature. Also, there was a significant rise in aural temperature following the test sessions.

Subjective alertness, sleepiness, and sleep quality

The mean values of these three variables in the nap and no-nap conditions are shown in Figure 2 (top). The nap condition was associated with a significantly increased alertness (*t*-tests: $t_9 = 5.77$, $P < 0.0005$) and decreased sleepiness ($t_9 = 6.74$, $P < 0.0005$). Nine of the 10 participants estimated that they obtained at least some sleep during the time allocated to a nap. Sleep quality was subjectively given a mean score of 4.20 ($s_{\bar{x}} = 0.79$), which is just below a score of 5, "some sleep with some interruptions". There were no significant correlations (Pearson correlation coefficient, *r*) between subjective sleep quality and the rise in alertness or fall in sleepiness ($P > 0.10$ in both cases).

Results from the test sessions

Visual short-term memory test

The results of the short-term memory tests are shown in Figure 2 (bottom) and the associated statistics in Table II (two-way ANOVA with repeated measures: dependent variable = score; fixed factors = nap/no-nap and digits-forwards/digits-backwards components). There were significant positive effects of the naps on both memory tests, and a tendency for the scores in the forward case to be higher than those in the backward case. Effects due to the nap did not correlate significantly with increases in alertness or falls in sleepiness with either the digits-forwards or digits-backwards components ($P > 0.2$ in all cases). Effects due to the nap did not correlate significantly with subjective sleep quality in the forwards case ($P > 0.7$); however, there was a significant negative correlation between the subjective sleep quality and the improvement after a nap in the digits-backwards case (Pearson's $r = -0.88$, $P = 0.001$).

Mean reaction times and reaction time errors

Figure 3 (top) and Table II show the results for the mean reaction times (the ANOVA was performed in the same way as for the memory tests, above). There was a highly significant effect of number of options on reaction time, but there was no significant difference between the nap and no-nap conditions. There was a significant positive correlation (Pearson's $r = +0.71$, $P = 0.021$) between the increase in alertness and the decrease in reaction time for the one-choice case; otherwise, there were no significant correlations between the effects of naps on changed reaction times and changes in alertness, sleepiness or subjective sleep quality.

The numbers of errors in the reaction time tests are shown in Figure 3 (bottom) and the associated statistics in Table II. Again, the number of choices had a significant effect on the number of errors. There was some evidence for an effect of the naps insofar as the interaction between nap/no-nap and number of choices reached $P = 0.085$ (see Table II). A specific comparison between errors in the nap and no-nap condition when there were eight choices confirmed the presence of a marginally significant difference (*t*-test: $t_9 = 2.06$, $P = 0.070$), as there were fewer errors in the nap condition. None of the differences in errors between the nap and no-nap conditions correlated with changes in alertness or sleepiness, or with subjective sleep quality.

Table I. Results of repeated-measures ANOVA for heart rate and aural temperature.

	Factor								
	Nap/no nap			Occasion of measurement			Interaction		
Variable	*F*	d.f.	*P*	*F*	d.f.	*P*	*F*	d.f.	*P*
Heart rate (pre-/post-nap)	**10.5**	**1, 9**	**0.01**	**22.8**	**1, 9**	**0.001**	**54.9**	**1, 9**	**<0.0005**
Heart rate (pre-/post-test session)	**8.0**	**1, 9**	**0.02**	**11.0**	**1, 9**	**0.009**	0.4	1, 9	0.53
Temperature (pre-/post-nap)	0.8	1, 9	0.41	0.3	1, 9	0.86	**8.8**	**1, 9**	**0.016**
Temperature (pre-/post-test session)	0.2	1, 9	0.66	**7.5**	**1, 9**	**0.023**	0.1	1, 9	0.93

Note: Significant results are shown in **bold** font.

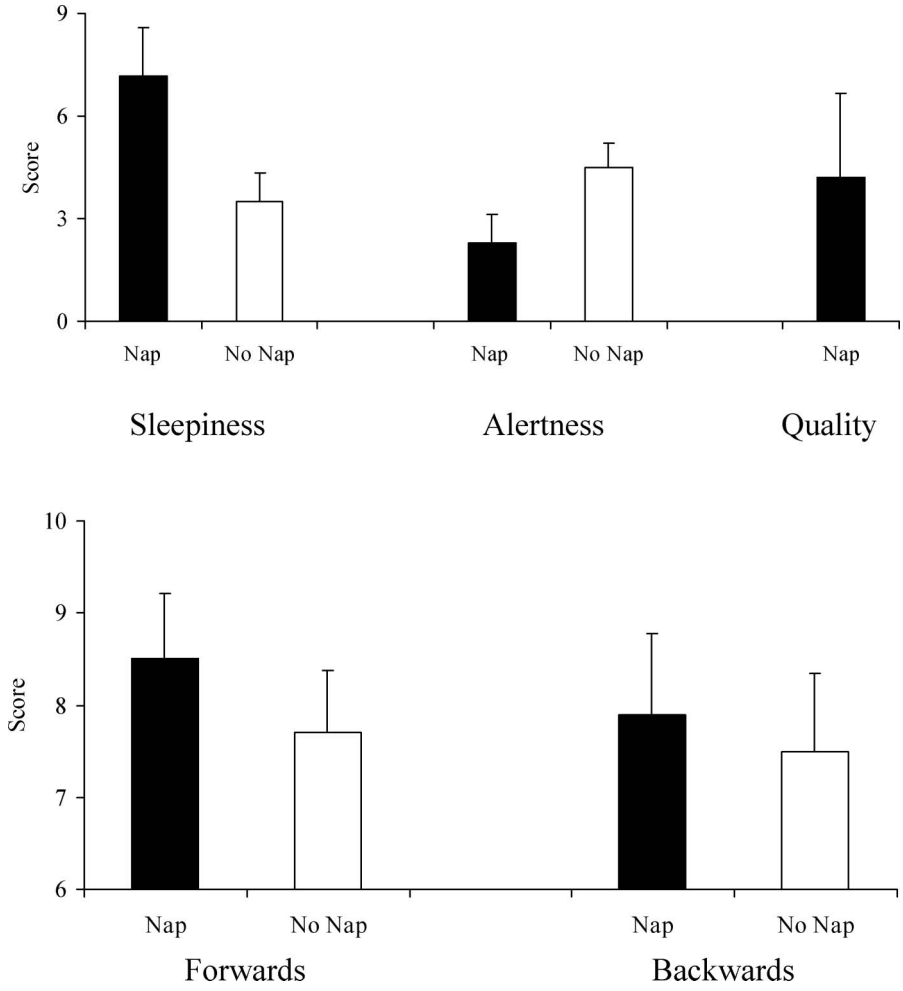

Figure 2. (Top) Subjective assessments of sleepiness and alertness in the nap and no-nap conditions, and of sleep quality. (Bottom) Scores for the short-term memory tests in the nap and no-nap conditions. Values are means and standard deviations. Forwards = digits-forwards component; Backwards = digits-backwards component. For more details, see text.

Table II. Results of repeated-measures ANOVA for short-term memory, mean reaction times, reaction time errors, and mean times for the 2-m and 20-m sprints.

		Factor 1 (see column to the left)			Factor 2 Nap/no nap			Interaction		
Variable	Factor 1	*F*	d.f.	*P*	*F*	d.f.	*P*	*F*	d.f.	*P*
Short-term memory	Forwards/backwards	*4.2*	*1.0, 9.0*	*0.07*	**9.5**	**1.0, 9.0**	**0.013**	1.4	1.0, 9.0	0.27
Reaction times – means	1, 2, 4, 8 choices	**93.1**	**1.4, 12.3**	**<0.0005**	0.1	1.0, 9.0	0.75	0.3	1.7, 15.1	0.68
Reaction times – errors	1, 2, 4, 8 choices	**18.0**	**1.3, 11.7**	**0.001**	2.5	1.0, 9.0	0.15	*3.0*	*1.7, 15.5*	*0.085*
2-m sprints – means	Sprints 1 – 3	*2.9*	*1.6, 14.7*	*0.094*	**6.8**	**1.0, 9.0**	**0.029**	1.4	1.4, 12.9	0.28
20-m sprints – means	Sprints 1 – 3	**4.4**	**1.6, 14.3**	**0.04**	**9.7**	**1.0, 9.0**	**0.013**	0.5	1.3, 11.6	0.55

Note: Significant results shown in **bold** font; $0.10 > P > 0.05$ shown in *italic* font.

Grip strength

Three-way ANOVA with repeated measures was used (dependent variable = grip strength; fixed variables = left/right hand, nap/no-nap, test number 1 – 3). The results (Table III) indicate no significant effect of the nap but a significant effect of hand (right stronger than left) and a weak effect of test number (test 1 giving the highest values). There were no significant interactions between the main factors. Differences in grip strength between the nap and no-nap conditions did not correlate significantly with changes in alertness and sleepiness, or with subjective sleep quality.

Sprint performance

Figure 4 shows the results for the 2-m and 20-m sprints. The associated statistics are shown in

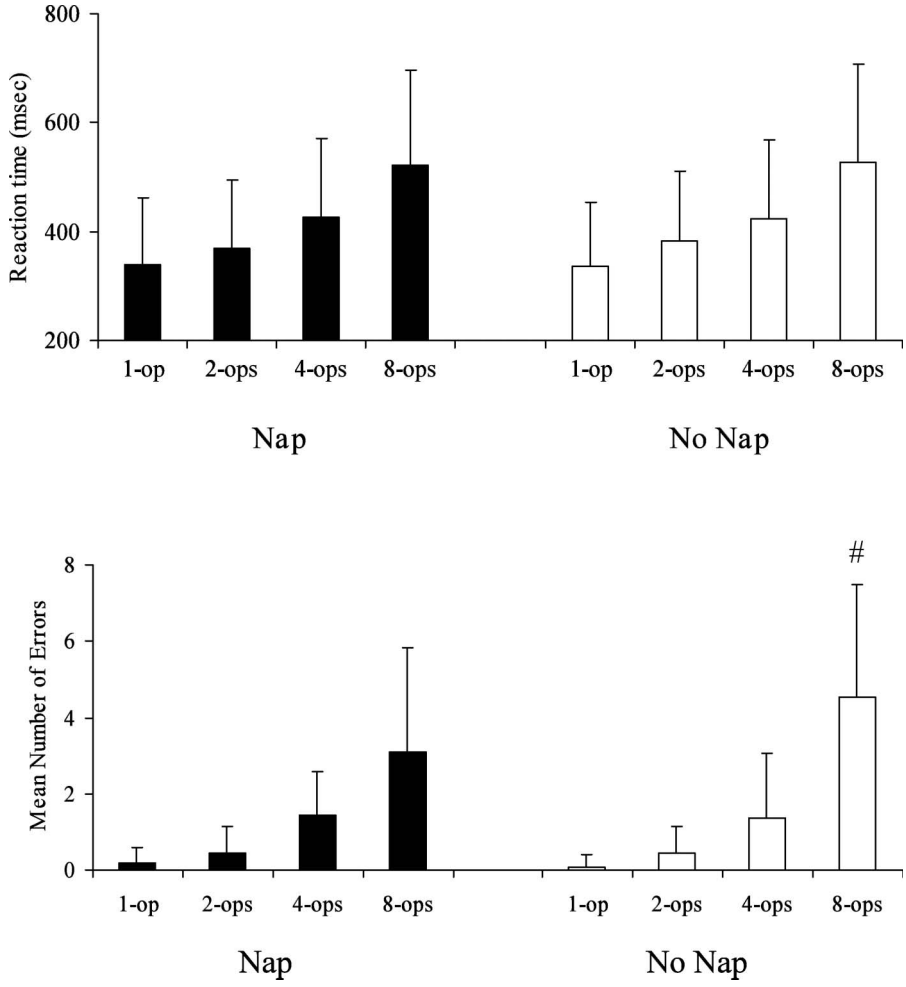

Figure 3. (Top) Reaction times for the one-, two-, four-, and eight-choice reaction time tests (1-op, 2-op, 4-op, and 8-op respectively) in the nap and no-nap conditions. (Bottom) Numbers of errors for the one-, two-, four-, and eight-choice reaction time tests in the nap and no-nap conditions. Values are means and standard deviations. $^{\#}P = 0.70$ (paired *t*-test).

Table III. Results of repeated-measures ANOVA for handgrip strength.

	F	d.f.	*P*
Main factors			
Left/right hand	**7.5**	**1.0, 9.0**	**0.023**
Nap/no nap	0.1	1.0, 9.0	0.80
Test number 1 – 3	*3.5*	*1.7, 15.4*	*0.063*
Interactions			
Hand × Nap	0.1	1.0, 9.0	0.93
Hand × Test number	1.7	1.6, 14.7	0.23
Nap × Test number	0.4	1.7, 15.5	0.65
Hand × Nap × Test number	1.4	1.2, 10.9	0.28

Note: Significant results shown in **bold** font; $0.10 > P > 0.05$ shown in *italic* font.

Table II (two-way ANOVAs with repeated measures; dependent variables = times for 2 m/20 m; fixed factors = nap/no-nap condition, sprint number 1 – 3). For both the 2-m and 20-m distances, there was a significant effect of nap. The mean time for the 2-m sprints fell from 1.060 s ($s_{\bar{x}} = 0.018$) to 1.019 s ($s_{\bar{x}} = 0.019$) ($P = 0.031$ paired *t*-test). The mean times for the 20-m sprints fell from 3.971 s ($s_{\bar{x}} = 0.054$) to 3.878 s ($s_{\bar{x}} = 0.047$) ($P = 0.013$). There was also a weak effect of sprint number, with the best times recorded during the first sprint; however, there were no significant differences ($P > 0.05$) when the mean times for sprints 1, 2, and 3 were individually compared.

The mean improvements in sprint performance at 2 m and 20 m correlated significantly with each other ($r = +0.69$, $P = 0.029$), but these improvements did not correlate significantly with rise in alertness (2 m: $r = +0.46$, $P = 0.18$; 20 m: $r = +0.36$, $P = 0.31$), fall in sleepiness (2 m: $r = +0.58$, $P = 0.08$; 20 m: $r = +0.39$, $P = 0.26$) or subjective sleep quality (2 m: $r = +0.31$, $P = 0.38$; 20 m: $r = -0.17$, $P = 0.64$).

The results from the study are summarized in Table IV.

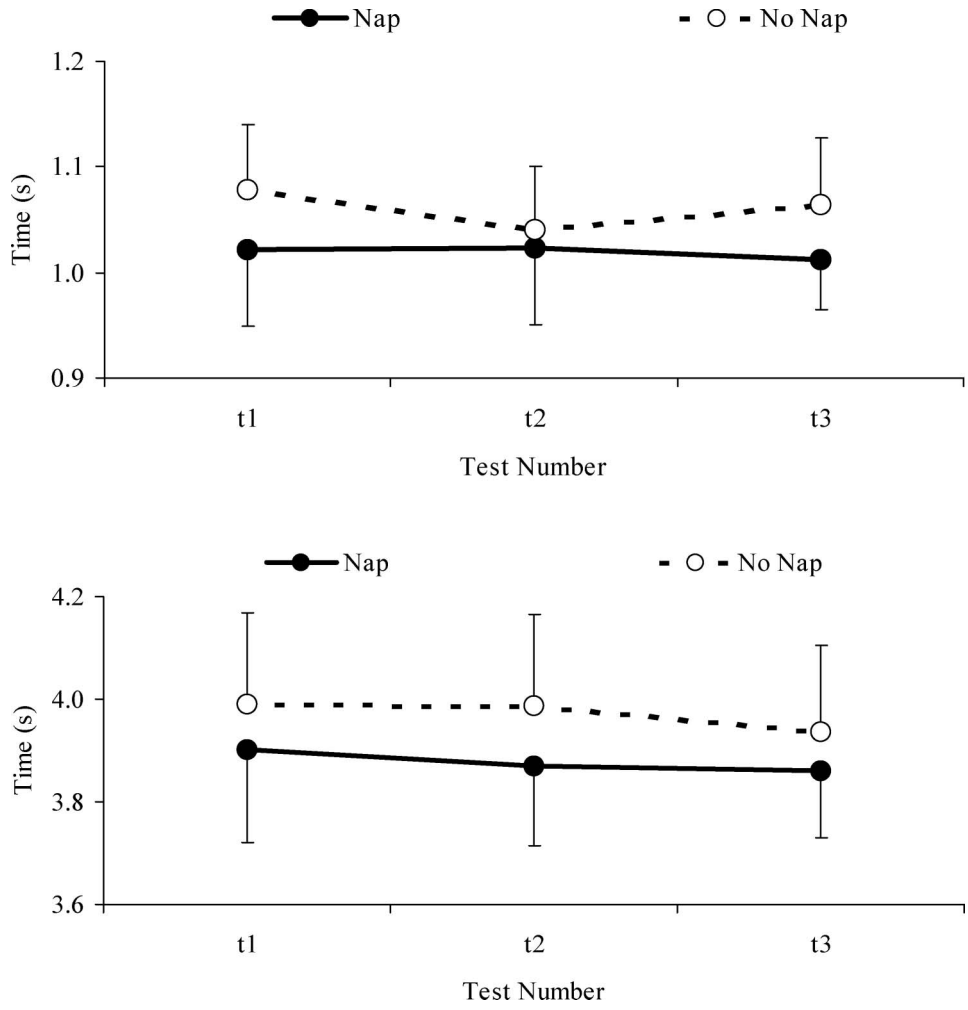

Figure 4. Times taken to cover 2 m (top) and 20 m (bottom) during the three sprints (t1 – t3) in the nap and no-nap conditions. Values are means and standard deviations.

Table IV. Summary of main significant results.

		Difference [Nap – no nap] condition correlated with subjective			
Variable	Effect of nap	Alertness	Sleepiness	Sleep quality	Other
Subjective alertness	↑			X	} Alertness and sleepiness
Subjective sleepiness	↓			X	} negatively correlated
Short-term memory	↑	X	X	X[a]	
Reaction times – mean	X	X[b]	X	X	1-choice < 2 < 4 < 8-choices
Reaction times – errors	X[c]	X	X	X	1-choice < 2 < 4 < 8-choices
Handgrip strength	X	X	X	X	Left < Right. Test 1 > Tests 2/3
2-m sprint time	↓	X	X	X	} Effects of nap on 2-m and 20-m
20-m sprint time	↓	X	X	X	} times are positively correlated

Note: ↑, rise; ↓, fall; X, not significant. Blank indicates not tested.
[a]Improvement of digits-backwards score correlated negatively with sleep quality.
[b]Improvement with nap in one-choice case correlated positively with increase in alertness.
[c]Evidence for decrease in errors in nap condition with the eight-choice case.

Discussion

The opportunity for a 30-min nap rather than sitting reading for the same period of time produced falls in heart rate and aural temperature (Figure 1 and Table 1), as well as a fall in sleepiness and rise in alertness (Figure 2, top). These changes would be anticipated (Reilly *et al.*, 1997). The lack of

correlations between the subjective assessments of sleep quality and changes in alertness and sleepiness (Table IV) suggests either that the subjective assessments were not internally consistent or that the value to alertness and sleepiness of the nap time was more in the relaxation than the quality of sleep that it provided.

Whatever the reason, it is clear that naps produced an improvement in performance at the sprints after sleep loss the previous night (Figure 4 and Tables II, IV). This improvement approached 0.1 s in the 20-m tests. We did not investigate whether this improvement is retained in races longer than 20 m; however, if this were the case, or if it were to be altered proportionally over a distance of 100 m, it would have a considerable influence on the individual's position in a competitive race. Any such improvement has a much wider application, however – it would also change the individual's satisfaction in training sessions, for example, and so be valuable on a daily basis.

The improvement in sprint performance following a nap was not matched by changes to grip strength (Tables III, IV). Also, even though grip strength and sprint times both showed small differences when the sequence of the three tests was considered, the directions of change were opposite to one another. For the sprints, performance tended to increase on successive occasions (Figure 4 and Table II); for grip strength, it tended to decline (Table III). The decline in grip strength is consonant with the concept of muscle fatigue, whereas the changes in sprint times cannot be explained by this factor but suggest some degree of ''warm-up'' instead.

That is, measuring grip strength (often used as a surrogate for sports performance) was not a substitute for measuring actual performance (sprint times) in this study. Whether other aspects of physical performance (e.g. aerobic power, isometric and isotonic muscle actions, and spine flexibility) would reflect the changes observed in the sprints is unclear. The observation that many aspects of physical performance have circadian rhythms that are synchronous with one another and sports performance might reflect no more than the fact that the sleep – wake cycle affects all these rhythms (Reilly *et al.*, 1997; Waterhouse *et al.*, 2005). In other words, how closely aspects of physical performance act as markers for sports performance is something that requires further investigation.

This difficulty in assessing sports performance from measures of physical performance is found also when ''mental performance'' is considered. Many tasks of mental performance show lower values in the early morning and late evening than in the middle of the waking day – but their detailed profiles vary, they show different susceptibilities to effects of time awake and sleep loss, and different recovery rates following naps. As a result, their ability to predict the effects of these factors on complex mental performance in field conditions can be limited (see, for example, Dinges, 1995; Folkard, 1990; Van Dongen and Dinges, 2005; Waterhouse *et al.*, 2001). The current results illustrate some of these differences between markers of mental performance. On the one hand, naps improved alertness and decreased sleepiness, both the digit-forwards and digit-backwards components of short-term memory improved in the nap condition (Figure 2 and Table II), and there was some evidence that naps decreased the error rate with the eight-choice reaction time test (Figure 3, bottom, and Tables II, IV). On the other hand, mean reaction times with the one-, two-, four-, and eight-choice versions of the reaction time tests (Figure 3, top, and Table II) were unchanged between the nap and no-nap conditions. It is also possible that the test of visual reaction time used in this study was insufficient in duration to detect the effects of sleep loss.

Even though subjective alertness and sleepiness, short-term memory, and sprint times all improved in the nap condition, there was no evidence of a close association between the variables. Changes in short-term memory and sprint times produced by the naps rarely correlated significantly with the changes in subjective alertness and sleepiness or subjective sleep quality (Table IV). Any link between these variables must be indirect, therefore, and its nature remains obscure.

The nap was taken around lunchtime, since this is one of the times when individuals, particularly those with some degree of sleep deprivation, show a transient fall in alertness and can initiate sleep more readily (Brooks & Lack, 2005; Reilly *et al.*, 1997), and when it can be incorporated into a daily routine. In the current study, 30 min was allowed after the nap and before the test sessions began, and this would have enabled any effects of sleep inertia that might have been present to have dissipated (Brooks & Lack, 2005; Naitoh, 1981; Stampi *et al.*, 1990). The changes produced by the nap were generally improvements, and this argues against their having been due to sleep inertia. Nevertheless, the observation that the amount of improvement in the digit-backwards component of the short-term memory test correlated *negatively* with subjective sleep quality (Table IV) fits with the view that residual effects of sleep inertia were present in this case. (The short-term memory test was performed first in the test sessions, see ''Methods''.) However, the absence of a negative effect of subjective sleep quality on performance change in the digit-forwards component, which was performed before the digit-backwards component, undermines this explanation and so the reason for the result is unclear.

There were several limitations to the present study that require further work to be performed. This work

could include an increased number of participants and objective measurement of sleep (to assess if sleep itself is required or if resting quietly or dozing is equally effective). It would also be instructive to investigate if nights of restricted sleep length altered the quality of sleep that was obtained.

Nevertheless, from a sports performance perspective, we have shown that a short nap has a beneficial effect on sprint performance in participants who have been partially deprived of sleep. We suggest that napping is of potential benefit to athletes during training and in a competitive environment. However, it remains to be determined if 30 min is the best length of time for a nap, whether taking the nap at different times of the day would be equally or more effective, and whether the improvement observed in the short sprints applies to longer track events and to a wider range of sporting activities.

References

Åkerstedt, T., & Folkard, S. (1995). Validation of the S and C components of the three-process model of alertness regulation. *Sleep*, *18*, 1–6.

Atkinson, G. (2001). Analysis of repeated measurements in physical therapy research. *Physical Therapy in Sport*, *2*, 194–208.

Atkinson, G. (2002). Analysis of repeated measurements in physical therapy research: Multiple comparisons amongst level means and multi-factorial designs. *Physical Therapy in Sport*, *3*, 191–203.

Brooks, A., & Lack, L. (2005). Naps. In C. Kushida (Ed.), *Sleep deprivation: Clinical issues, pharmacology, and sleep loss effects* (pp. 457–474). New York: Marcel Dekker.

Buckolz, E., & Vigars, B. (1987). Sprint start reaction time: On the advisability of sensory vs. motor sets. *Canadian Journal of Sports Science*, *12*, 51–53.

Dinges, D. (1995). An overview of sleepiness and accidents. *Journal of Sleep Research*, *4* (suppl. 2), 4–14.

Folkard, S. (1990). Circadian performance rhythms: Some practical and theoretical implications. *Philosophical Transactions of the Royal Society of London B*, *327*, 543–553.

Hick, W. (1952). On the rate of gain of information. *Quarterly Journal of Experimental Psychology*, *4*, 11–26.

Hoddes, E., Zarcone, V., Smythe, H., Phillips, R., & Dement, W. (1973). Quantification of sleepiness: A new approach. *Psychophysiology*, *10*, 431–436.

Krauchi, K., Cajochen, C., & Wirz-Justice, A. (2005). Thermophysiologic aspects of the three-process model of sleepiness regulation. *Clinics in Sports Medicine*, *24*, 287–300.

Leger, D., Metlaine, A., & Choudat, D. (2005). Insomnia and sleep disruption: Relevance for athletic performance. *Clinics in Sports Medicine*, *24*, 269–285.

Meney, I., Waterhouse, J., Atkinson, G., Reilly, T., & Davenne, D. (1998). The effect of one night's sleep deprivation on temperature, mood, and physical performance in subjects with different amounts of habitual physical activity. *Chronobiology International*, *14*, 125–132.

Naitoh, P. (1981). Circadian cycles and restorative power of naps. In L. Johnson, D. Tepas, P. Colquhoun, & M. Colligan (Eds.), *Biological rhythms, sleep and shift work* (pp. 553–580). New York: Spectrum.

Neri, D. (2004). Proceedings of the Fatigue and Performance Modeling Workshop, 13–14 June 2002, Seattle, WA. *Aviation, Space, and Environmental Medicine*, *75*, Section II (suppl.), A1–A199.

Reilly, T., Atkinson, G., & Waterhouse, J. (1997). *Biological rhythms and exercise.* Oxford: Oxford University Press.

Reilly, T., Edwards, B., Waterhouse, J., & Atkinson, G. (2005). Jet lag and air travel: Implications for performance. *Clinics in Sports Medicine*, *24*, 367–380.

Reilly, T., & Walsh, T. (1981). Physiological, psychological and performance measures during an endurance record for 5-a-side soccer play. *British Journal of Sports Medicine*, *15*, 122–129.

Rosekind, M., Flower, D., Gregory, K., & Jung, W. (2005). General occupational implications of round-the-clock operations. In C. Kushida (Ed.), *Sleep deprivation: Clinical issues, pharmacology, and sleep loss effects* (pp. 229–249). New York: Marcel Dekker.

Smith, R., & Reilly, T. (2005). Athletic performance. In C. Kushida (Ed.), *Sleep deprivation: Clinical issues, pharmacology, and sleep loss effects* (pp. 313–334). New York: Marcel Dekker.

Spencer, M. (1987). The influence of irregularity of rest and activity on performance: A model based on time since sleep and time of day. *Ergonomics*, *30*, 1275–1276.

Stampi, C., Mullington, J., Rivers, M., Campos, J., & Broughton, R. (1990). Ultrashort sleep schedules: Sleep architecture and recuperative value of 80-, 50- and 20-min naps. In J. Horne (Ed.), Sleep '90 (pp. 71–74). Bochum: Pontenagel Press.

Troy, S., Lucki, I., Unruh, M., Cevallos, W., Leister, C., Martin, P. *et al.* (2000). Comparison of the effects of Zalepon, Zolpidem and Triazolam on memory, learning and psychomotor performance. *Journal of Clinical Psychopharmacology*, *30*, 328–337.

Van Dongen, H., & Dinges, D. (2005). Sleep, circadian rhythms, and psychomotor vigilance. *Clinics in Sports Medicine*, *24*, 237–249.

Waterhouse, J., Drust, B., Weinert, D., Edwards, B., Gregson, W., Atkinson, G. *et al.* (2005). The circadian rhythm of core temperature: Origin and some implications for exercise performance. *Chronobiology International*, *22*, 207–225.

Waterhouse, J., Minors, D., Åkerstedt, T., Reilly, T., & Atkinson, G. (2001). Rhythms of human performance. In J. Takahashi, F. Turek, & R. Moore (Eds.), *Handbook of behavioral neurobiology: Circadian clocks* (pp. 571–601). New York: Kluver Academic/Plenum Publishers.

Zar, J. (1999). Biostatistical Analysis. London, Prentice.

Copyright of Journal of Sports Sciences is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.